Statistical Theory
Analysis on Road Accidents

Gilad Mizrahi Noa Kohn

***Abstract*—In this study, we analyzed 2024 road accident data from Data.gov.il to examine how road type, time of day, weather, and accident type relate to severity. After data cleaning, the sample was filtered from 8,315 to 6,274 accidents. Chi-square tests revealed significant relationships between accident severity and road type, time of day, and accident type, while weather conditions alone were non-significant. A logistic regression model was constructed to predict minor versus severe/fatal accidents, incorporating main effects and interactions for road type with both accident type and weather. The adjusted model fit the data significantly better than a null model ($p < 0.001$), yielding a McFadden Pseudo-R^2 of 0.071 and a moderate AUC of 0.664. The regression confirmed that the impact of accident type (e.g., pedestrian involvement) and weather on severity varies significantly across different road types. In conclusion, while these environmental and situational factors contribute to understanding accident severity, they are insufficient for highly accurate prediction. Future models should incorporate additional predictors such as speed, driver characteristics, or traffic volume.**

***Index Terms*—Logistic regression, accident severity analysis, road accidents.**

I. Introduction

Road accidents are a major cause of injuries and fatalities in Israel and around the world. Therefore, it is important to understand which characteristics are associated with a higher risk of severe injury or death.

Previous studies have extensively examined factors related to accident severity; for instance, a 2024 study analyzing pedestrian-vehicle accidents in Berlin found that vehicle type, lighting conditions, and vehicle speed were associated with pedestrian injury severity [1]. In addition, a study published in 2019, based on night-time vehicle accidents under low illumination conditions in a city in China during 2014–2016, found that accident location, accident type, and the presence of a median divider were associated with accident severity [2].

The current study examines the relationship between road type, time of accident, weather conditions, and accident type and accident severity in road accidents with casualties in Israel in 2024. [3]

To ensure reproducibility, the complete R source code utilized for data cleaning, modeling, and visualization is publicly available at the GitHub repository https://github.com/NotUniqueUser/stat-theory-project.

II. Results

A. Data Import and Cleaning

Our raw data contains 8,315 cases of reported road accidents in 2024. However, in some cases there are missing information or is simply tagged "Unknown". We omitted the following rows:

- **Weather condition** – rows where the weather condition for the accident were unknown.
- **Time of event** – rows where the time of the accident (day/night) was omitted.
- **Type of accident** – rows where the accident was marked as "other".
- **Road type** – rows where the type of road was omitted.
- **Coordinates** – rows where the (X, Y) coordinates of the accidents were omitted.

The sample size was reduced from 8,315 to 6,274 accidents.

B. Visualization of the Data

To gain a grasp of the distribution of accidents across the country, we first plot the accidents over the geographical map of Israel and mark points depending on the severity of the accident (only severe and fatal brevity).

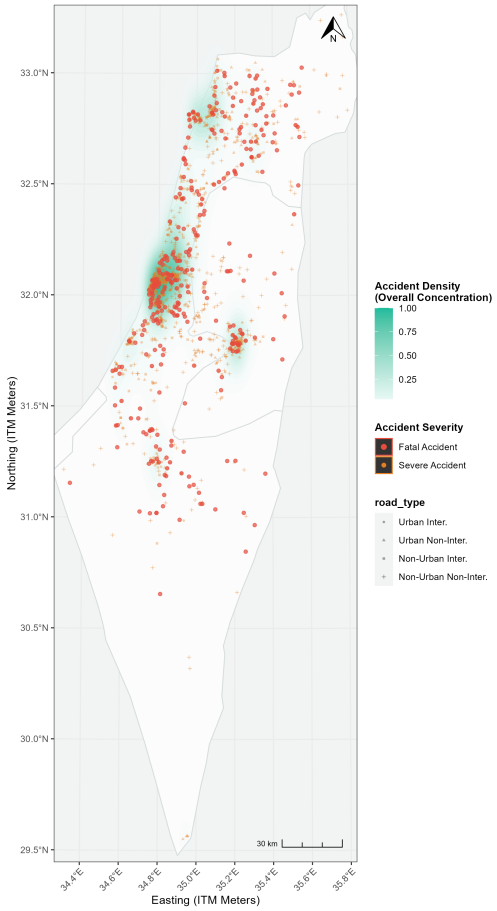


Fig. 1: Overlay of Fatal/Severe events on overall accident density

We can already see that major cities (Tel-Aviv, Jerusalem, Haifa) contain a large amount of the overall accidents. For a quantitative view, we also plot vertical bar charts to show the number of accidents in each road type and their share of severity

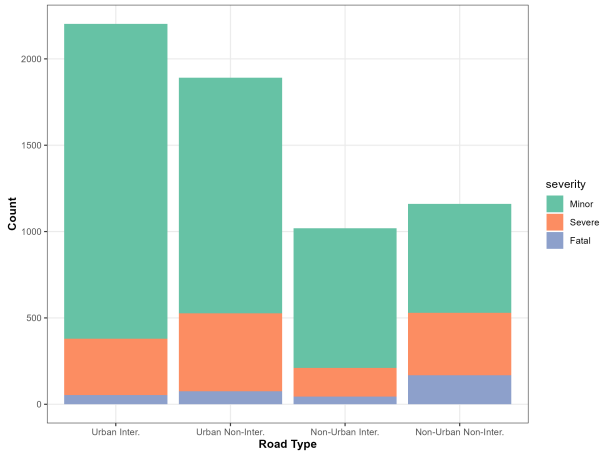


Fig. 2: Number of accidents by severity and road types

An interesting observation is that despite urban areas experiencing over twice the number of accidents accidents, they're usually much less lethal than the non-urban non-intersection accidents, which carry a considerable amount of fatalities.

C. *Features, relations and χ^2 tests*

We now have the following features for each accident:

- **Weather condition** – whether it was clear or rainy/-foggy.
- **Time of event** – whether the accident occured at day or night
- **Type of accident** – goes into 3 categories:
- **Road type** – Urban/Non-Urban and Intersection/Non-Intersection
  - **Pedestrian** – a vehicle that hit a pedestrian.
  - **Car crash** – a vehicle hitting another car in any way (front, back, side).
  - **Self crash** – a vehicle colliding with a still object or rolling off the road (self accident).
- **Road state** – whether the road was dry or wet.
- **Severity** – 3 categories:
  - **Minor** – at least one person sustained minor injuries.
  - **Severe** – at least one person sustained severe injuries.
  - **Fatal** – at least one person died as a result of the accident.

Additionally we make use of a feature of our own: **Is severe/fatal** – to make a binary distinction between accidents that carry long-term issues (death, chronic pain...).

Since our features are nominal, we can not rely on a correlation matrix to check the association between features. We use a Cramér's V matrix [4] to initially spot any variables that could lead to multicollinearity later.

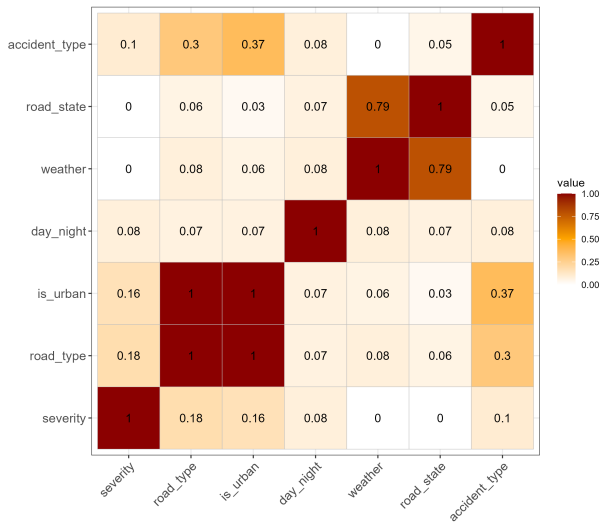


Fig. 3: Cramér's V correlation matrix

A non-trivial noticeable relation is between our weather and road state features. This can make sense intuitively, since a rainy day will almost surely result in a wet road. We construct Pearson's χ^2 tests between variables to check for dependence, and adjust their p-values using the Holm-Bonferroni method to avoid potentially large type I errors:

Feature #1	Feature #2	p-value	*adj.* p-value
Road type	Severity	7.36×10^{-73}	2.94×10^{-72}
Time of event	Severity	3.63×10^{-6}	7.26×10^{-6}
Weather	Severity	0.927	0.927
Accident type	Severity	2.99×10^{-28}	8.99×10^{-28}
Road state	Weather	< 0.001[1]	< 0.001[1]

TABLE I: χ^2 tests between features and their p-values before and after adjustment

We notice that our suspicion was correct: there is a very strong association with our road state and weather features. We opt to omit the road state feature from our model in the next section.

As for the other features, we find that there is indeed a dependency between the severity of an accident and conditions such as the time, the road type, and the accident type. This is intuitive: realistically, a pedestrian collision can result in a much more deadly outcome than a driver who has accidentally collided with a tree (thanks to safety mechanism such as the airbag)

Despite the insignificant result between our weather and severity features, we will still include it in our model to show how this might change when paired with other features.

[1]The p-value was so minuscule that the software we used could not estimate it and gave us a result of 0

D. Logistic Regression Specifications, Odd Ratios and Results

We will use the following logistic regression model with 80% of our data (split for later testing) to model our **Is severe/fatal** feature:

$$\begin{aligned}\ln\left(\frac{p}{1-p}\right) = \beta_0 &+ \beta_1 X_{\text{road}} + \beta_2 X_{\text{time}} \\ &+ \beta_3 X_{\text{weather}} + \beta_4 X_{\text{type}} \\ &+ \beta_5 (X_{\text{road}} \times X_{\text{type}}) \\ &+ \beta_6 (X_{\text{road}} \times X_{\text{weather}})\end{aligned} \tag{1}$$

After fitting the weights, we extract the model odds ratios with a 95% confidence interval.4

Some key findings are:

- **Severity of non-urban inter. and pedestrians accidents** – both individually and combined, they are significantly more deadly than other types of road and accident types respectively. When combined together, we can intuitively think about the fact that a pedestrian accident which would occur on an non-urban intersection would require paramedics to travel a longer distance than a similar accident in an urban area, unfortunately reducing the likelihood of survival.
- **Non-urban non-inter. roads** – with an odds ratio value higher than 3.0, we see that just according to our vertical bar plot2, accident on this type of road are significantly more lethal than ones on urban inter.
- **Time of day** – we observe that accidents during nighttime are significantly more lethal than during daytime.
- **Weather** – even though we observed independence on our previous test, we can still, surprisingly, see a significant negative correlation: during bad weather conditions, severe and fatal accidents are less likely. One hypothesis to this phenomenon is that drivers will try driving more carefully and slowly during such times.

E. Logistic Regression Model's Accuracy

We use several metrics to check our model's predictive power and overall accuracy:

1) **Comparison to a null model** – we take

$$\ln\left(\frac{p}{1-p}\right) = \beta_0$$

and run a Likelihood Ratio Test (LRT) with our fitted model. This is less of an accuracy test and more of "sanity test" to see if our model is any better than nothing but a model whose predictive

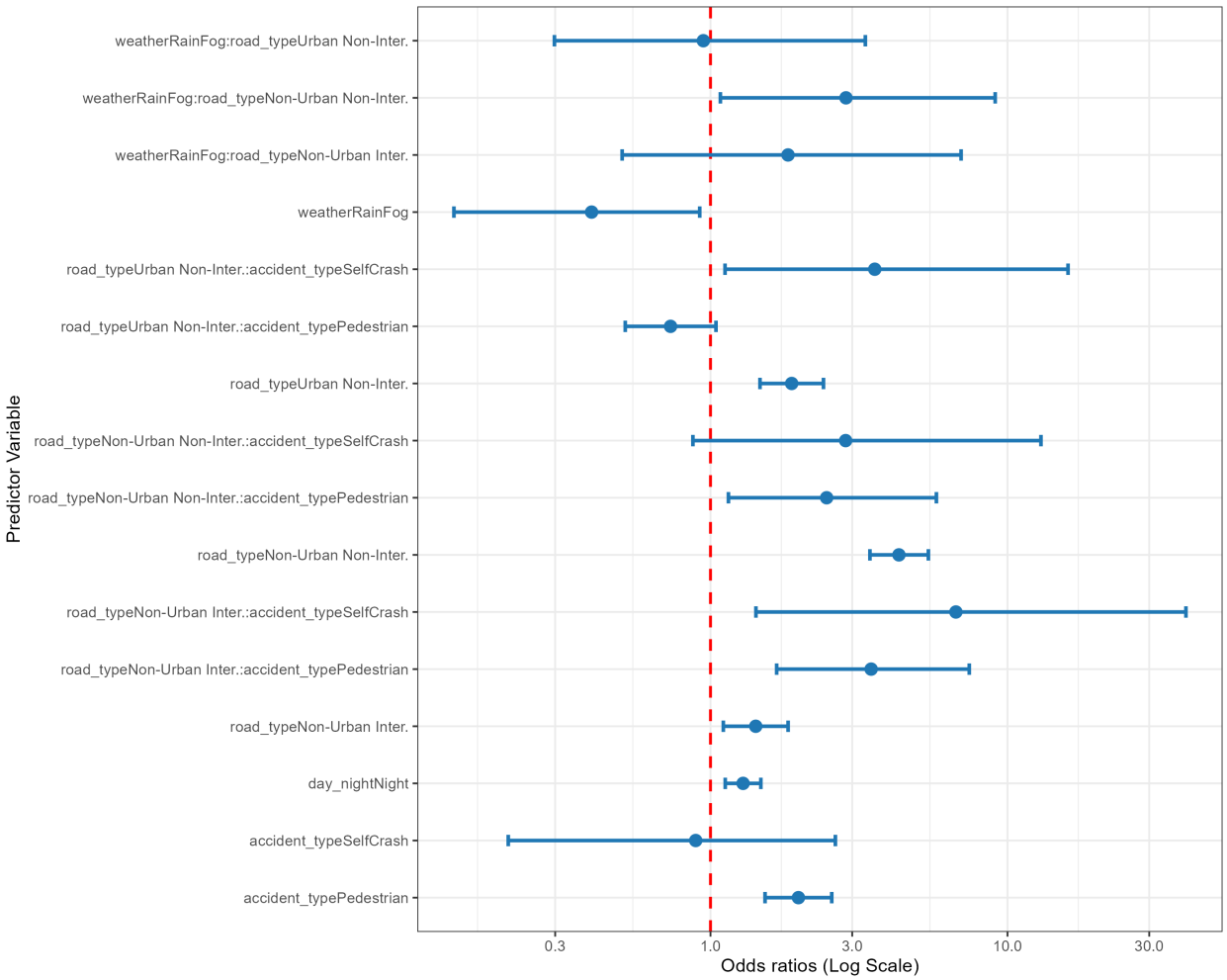


Fig. 4: Model Odds Ratios with 95% Confidence Intervals

power is only by the ratios of severe/fatal accidents. We get a p-value of < 0.001 as expected.

2) **McFadden Pseudo-R^2** – a metric for goodness-of-fit in logistic regression, we get a value of $R^2_L = 0.071$. While this value appears low, it is typical and acceptable for large-scale, observational macro data.
3) **ROC-AUC** – to measure predictive power, we plot a ROC curve for our model using the remaining 20% of data we put aside and calculate the area under the curve (AUC).

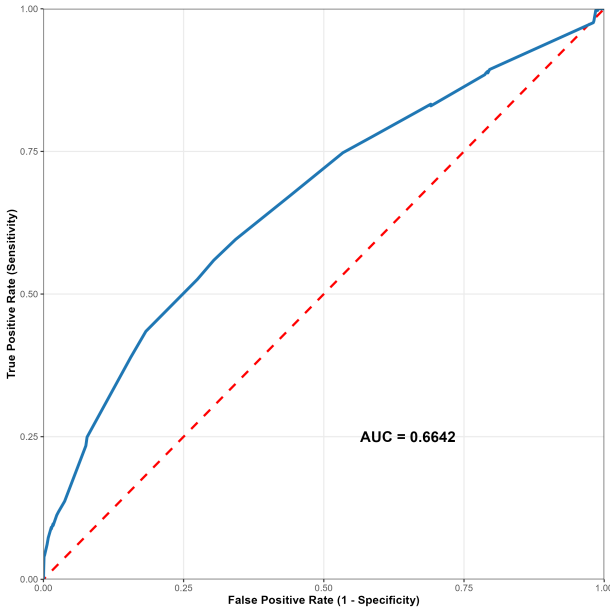


Fig. 5: ROC Curve for our Accident Severity Predictor Model

The AUC is ≈ 0.664, which is a mediocre but much better than 0.5 (predictive power of a coin flip).

III. Methods

A. Pearson's χ^2 test

A statistic test to test the independence of two features with r and k possible categories. The test hypothesis is

$$H_0 : \text{the features are independent}$$
$$H_1 : \text{the features are dependent}$$

Let us imagine a contingency table between those two variables where O_{ij} will be the number of observations where i and j occur for each respective feature. We then under the null hypothesis define

$$E_{ij} = \frac{R_i C_j}{N}$$

to be the expected frequency, where R_i, C_j, N are the total number of cases in row and column i and j and the total number of cases respectively. Then the χ^2 test statistic can be calculated

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{k} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

To get our p-value, we then calculate the right-side curve area between $\left[\chi^2, \infty\right]$ in a χ^2 distribution with $df = (r-1) \cdot (k-1)$.

B. Cramér's V

A measure of association between two variables which are not ordinal (nominal). We use the same exact conditions and χ^2 statistic in Pearson's χ^2 test. Then

$$V = \sqrt{\frac{\chi^2 / N}{\min\left(r-1, k-1\right)}}$$

An important condition is that we first must deny the null hypothesis for the corresponding Pearson χ^2 test of the relevant pair. In our plot, we zeroed out any pair that did not pass.

C. Holm-Bonferroni

An improvement over the standard Bonferroni correction to solve the multiple comparisons problem caused. We first sort our p-values by size such that

$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$$

where m is the total number of values. Instead of using a fixed significance level, we choose

$$\alpha_i = \frac{\alpha}{m - i + 1}$$

for each value on index i.

We then starting comparing from $i = 1$: if $p_{(1)} < \alpha_1$ we move to index 2 and so forth. We stop at the first test where $p_{(k)} \not< \alpha_k$, and then declare that all tests from 1 to $k-1$ are significant while k and onward are in-significant.

D. Logistic Regression

The fundamental classification algorithm that returns a probability for a binary target. We use a Sigmoid function

$$p = \sigma\left(z\right) = \frac{1}{1 + e^{-z}}$$

Where z is our linear equation of features with weights

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

When z becomes positive, we approach $p = 1$, and when negative we approach $p = 0$. The cutoff threshold occurs at $z = 0, p = 0.5$.

Note that:

- Solving the first equation for z yields

$$z = \ln\left(\frac{p}{1-p}\right)$$

  We call this the logit model formula.
- The weights β_i are fitted into the model using Maximum Likelihood Estimation (MLE)

E. Odd Ratio (OR)

OR is our way to measure the *relative change* in the odds of our outcome to occur, given that a single predictor value change.

Recall the formula from our logistic regression. These odds in log form. For OR we write

$$\mathrm{OR}_i = e^{\beta_i}$$

An $\mathrm{OR} = 1.0$ represents base neutrality (no effect on the odds). If $\mathrm{OR} > 1$ our odds of occurrence are increased by that number (e.g., $\mathrm{OR} = 1.3$ translates to a 30% increase in odds). Conversely, an OR lower than 1.0 represents an odds decrease in a similar manner.

In our analysis, we used a 95% confidence intervals of the weights we fitted in the logistic regression model to know our OR are significant.

F. Likelihood Ratio Test (LRT)

We use this test to evaluate whether our predictor variables improve the model's fit. Define the LRT statistic to be

$$\lambda_{\mathrm{LR}} = -2 \ln\left(\frac{L_0}{L_M}\right)$$

which is the difference between the deviance ($D = -2\ln(L)$) of our null model and full model.

We then utilize a χ^2 distribution with $df = k_M - k_0$, the difference in number of features between the models, and calculate $p = \Pr\left(\lambda_{\mathrm{LR}} \geq \chi^2_{k_M - k_0}\right)$.

G. McFadden Pseudo-R^2

This, alongside the LRT test, helps us *measure* how well we actually are from the base null model. The formula is

$$R_L^2 = 1 - \left(\frac{\ln L_0}{\ln L_M} \right)$$

H. Receiver Operating Characteristic (ROC) and Area Under Curve (AUC)

We use ROC to plot our model's performance based on a decision threshold. We let a fitted model predict results on data not seen before (test) and give it a decision thresholds (between $[0, 1]$). Each time, we use a confusion table and write

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \qquad \mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}$$

to be the true positive ratio and false positive ratio respectively. We then plot $(\mathrm{TPR}, \mathrm{FPR})$ each time. Note that for threshold 0 and 1 (predict all with 100% certainty) we'll get $(0, 1), (1, 0)$ respectively.

Now, to assess the model's discriminatory power, we calculate the Area Under the ROC Curve (AUC) and get a value between $[0.5, 1]$. A value of 0.5 signifies a completely useless model, with a predictive power that is as good as a coin flip.

IV. Discussion

The chi-square tests showed significant relationships between accident severity and road type, time of accident, and accident type. In contrast, no significant relationship was found between weather conditions and accident severity. In the logistic regression model, which included all variables together, it was found that the relationship between accident severity and accident type, as well as the relationship between accident severity and weather conditions, may vary across different road types.

However, the model has several limitations. Although it fit the data significantly better than a null model, the McFadden Pseudo-R^2 value obtained was 0.071. This means that the variables included in the model improve the fit to the data compared with a null model, but the improvement is not large. In addition, the AUC value obtained was 0.664, meaning that the model distinguishes between minor accidents and severe or fatal accidents better than random guessing, but its ability to distinguish between the groups is not high.

In addition, the analysis was based only on accidents reported in Israel in 2024 and included a limited number of variables. Additional variables that may affect accident severity, such as driving speed, driver characteristics, lighting conditions, traffic volume, and vehicle safety features, were not included in the model. Including these variables in future research may improve the ability to explain and predict accident severity.

References

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